Superstore Marketing Campaign Analysis

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BUSINESS OVERVIEW AND KPIs

The Superstore chain runs a year-end promotional campaign where existing customers are invited to purchase a Gold Membership that offers a 20% discount on all purchases. The goal of this project is to analyze customer data to understand who is likely to accept the membership, which customers spend the most, and how future campaigns can be improved.

Dataset Overview:

The raw dataset contained 2,240 customers and 22 original columns covering demographics, spending behavior, channel usage, and campaign response. After data cleaning and feature engineering in Python, 7 new columns were added — Age, Total Spend, Total Purchases, Tenure, Category Breadth, Web Ratio, and Income Bracket — bringing the final dataset to 29 columns.

Key Performance Indicators (KPIs):

KPI	Value
Total Customers	2,240
Gold Membership Response Rate	14.91%
Total Revenue Spent by Customers	$1,356,988
Average Spend per Customer	$605.80
Complaint Rate	Very low
Customer Enrollment Period	Jan 2012 to Dec 2014

What is Response Rate in simple words: Out of 2,240 customers, only 334 said YES to the Gold Membership. 334 divided by 2240 equals 14.91%. This means 85% of customers did NOT accept — there is a lot of room to improve.

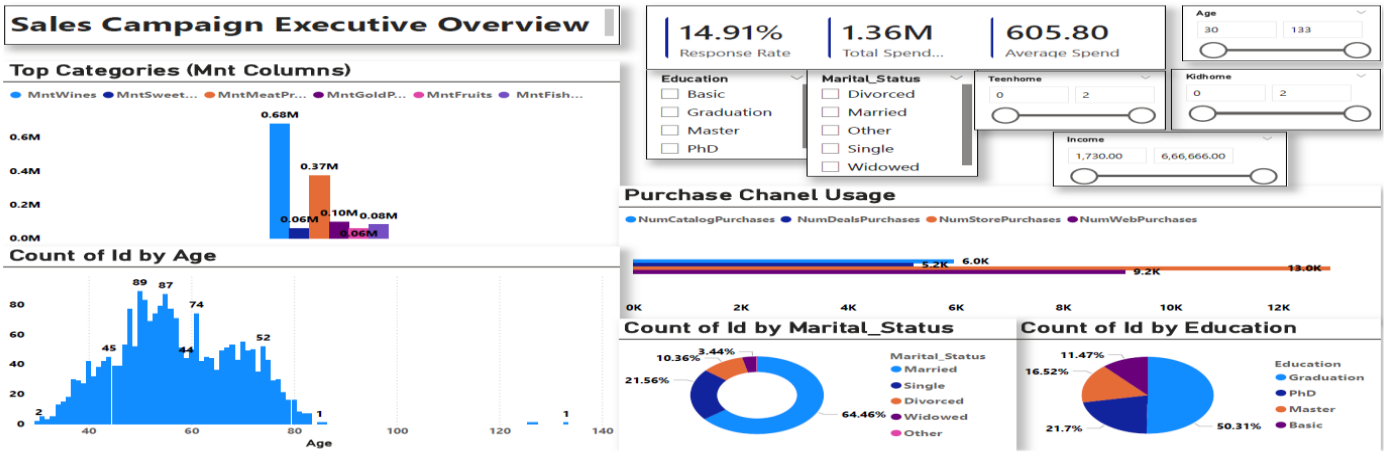

Figure 1: Executive Overview Dashboard showing Response Rate 14.91% and Total Spend

DATA ACQUISITION AND VALIDATION

The data was first imported into Excel for a visual check, then loaded into a MySQL database, and finally connected to Python using SQLAlchemy for advanced analysis.

Validation Results:

Total rows confirmed as 2,240 with no duplicate IDs. The Income column had 24 missing values which were filled using the median income value in Python. No duplicate customer records were found.

Data Anomalies Found:
One customer had a birth year of 1893, making their calculated age 133 years — this is clearly a data entry error. One customer had an income value of 666,666 — this is an extreme outlier far above all other income values. Both anomalies were noted and handled appropriately.
Date Information:
Customers enrolled between 8 January 2012 and 6 December 2014. The gap between the first and last enrollment date is 1,063 days, which means the company gathered its customer base over approximately 3 years.
Feature Engineering Done in Python:

New Column Added	How It Was Created
Age	2026 minus Year_Birth
Total_Spend	Sum of all 6 Mnt columns
Total_Purchases	Web + Catalog + Store purchases added
Tenure	Days since customer enrolled
Category_Breadth	Count of product categories with non-zero spend
Web_Ratio	Web purchases divided by Total purchases
Income_Bracket	Low, Medium, High groups created from Income

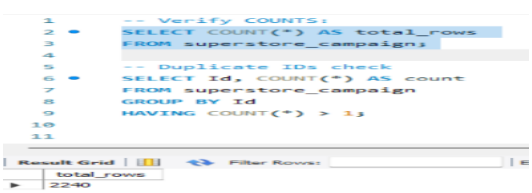

Figure 2:SQL Query confirming 2,240 rows loaded with no duplicates

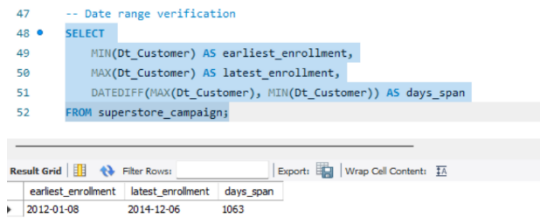

Figure 3: Enrollment date range — 2012 to 2014, spanning 1,063 days

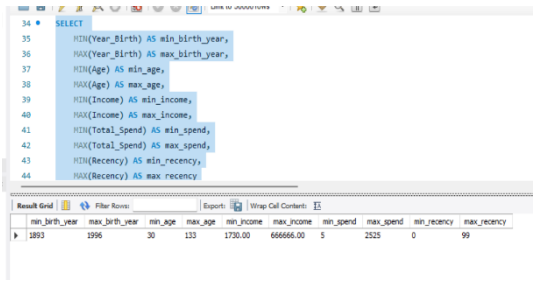

Figure 4: Range check showing age outlier of 133 and income outlier of 666,666

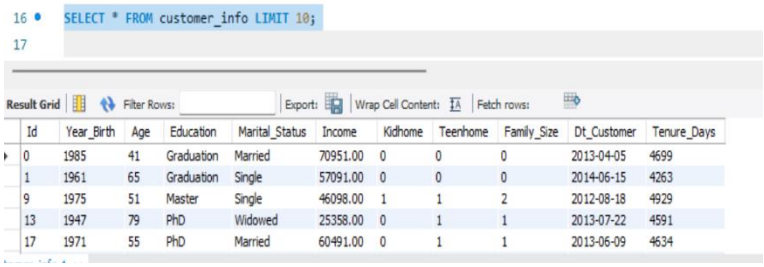

Figure 5: customer_info view showing Age, Tenure, and Family Size fields

EDA PART 1 — CUSTOMER DEMOGRAPHICS

Education Level: The majority of customers, 1,127 out of 2,240, are Graduates making up 50.31% of the base. PhD holders are second at 21.7% with 486 customers. Master's degree holders are 16.52% and Basic education customers are the smallest group at 11.47%.

Marital Status: Married customers dominate at 64.46% with 1,444 customers. Single customers are 21.56% with 483. Divorced customers are 10.36% with 232. Widowed customers are 3.44% with 77. A small group of 4 customers had unusual entries like "Absurd" which were grouped under "Other."

Age Distribution: Customer ages range from 30 to 99 years with one outlier at 133 (data error). The majority of customers are between 40 and 70 years old based on the histogram.

```
-- Education categories
SELECT Education, COUNT(*) AS count
FROM superstore_campaign
GROUP BY Education
ORDER BY count DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conten

Education	count
Graduation	1127
PhD	486
Master	370
Basic	257

Figure 6: Education category breakdown — Graduation is the largest group

```
-- Marital Status categories
SELECT Marital_Status, COUNT(*) AS count
FROM superstore_campaign
GROUP BY Marital_Status
ORDER BY count DESC;

-- Total spending across all categories
SELECT
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Marital_Status	count
Married	1444
Single	483
Divorced	232
Widowed	77
Other	4

Figure 7: Marital status distribution — Married customers are 64.46% of the base

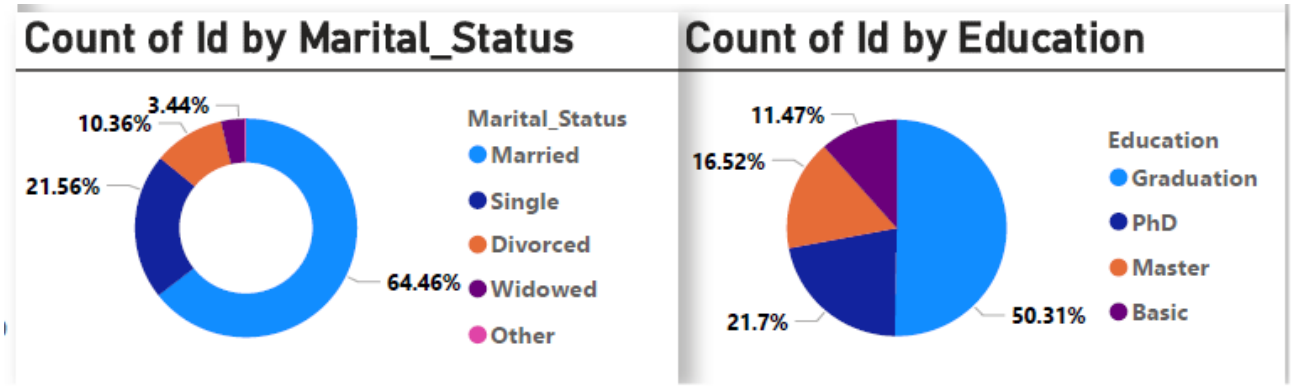

Figure 8: Visual breakdown of Education and Marital Status from Power BI dashboard

EDA PART 2 — SPENDING AND CHANNEL ANALYSIS

Spending by Product Category: Wine is the top revenue category with $680,000 in total customer spending. Meat products are second at $370,000. Gold products generated $100,000. Fruits and Sweets are the lowest performing categories at $60,000 each. Wine and Meat together represent approximately 77% of all customer spending.

Total Spend Distribution: The spending distribution is right-skewed. Most customers spend very little — over 650 customers spend close to zero. A small number of customers spend up to $2,500. This is a classic pattern where a small group of customers drives most revenue.

Income Distribution: The Income Boxplot shows that most customers earn between $30,000 and $90,000. There is one extreme outlier at $666,666 which was flagged as a data error.

Purchase Channel Usage: In-Store shopping is the most popular channel with 13,000 total store purchases. Web purchases total 9,200 and catalog purchases are 6,000. Deals purchases are 5,200.

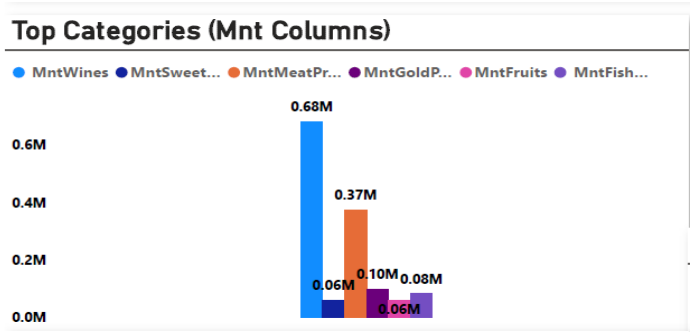

Figure 9: Total spend by product category — Wine leads at $0.68M

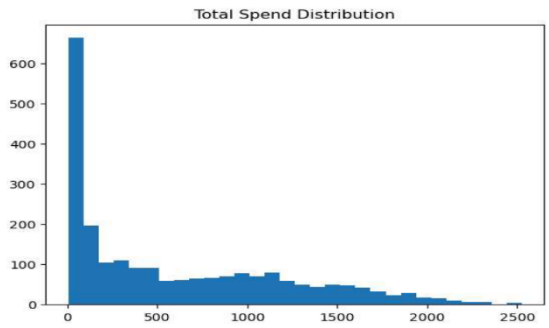

Figure 10: Right-skewed total spend distribution — most customers are low spenders

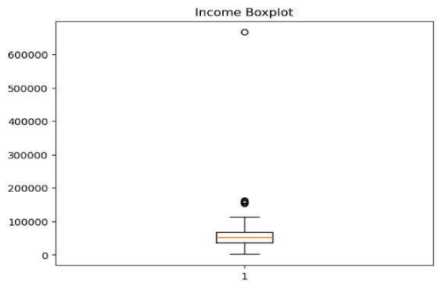

Figure 11: Income distribution with one extreme outlier at 666,666

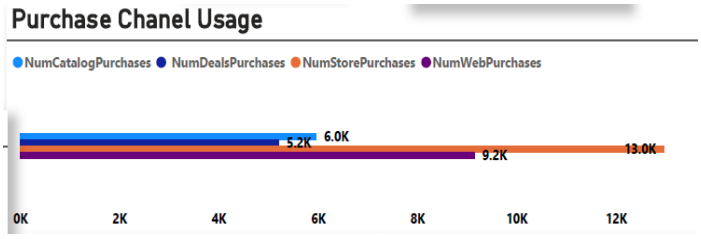

Figure 12: Purchase channel usage — In-store is dominant

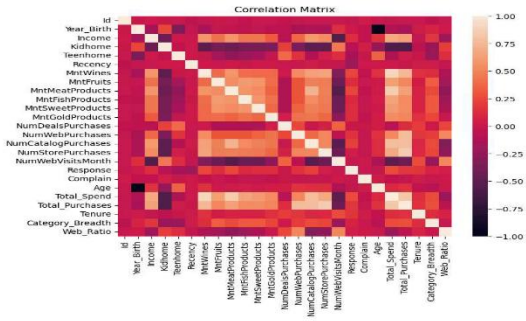

Figure13: Correlation matrix showing Income and Total Spend are strongly linked to Response

CUSTOMER SEGMENTATION

Segmentation by Spending Level:

Segment	Spend Range	Customer Count
Low Spender	Below $500	1,245 (55.6%)
Medium Spender	$500 to $1,500	749 (33.4%)
High Spender	Above $1,500	246 (11%)

The majority of customers are Low Spenders. However the 246 High Spenders are the most valuable group and should receive priority attention.

Segmentation by Shopping Channel:

Segment	**Count**
In-Store Shopper	1,563
Digital Shopper	370
Mixed Shopper	297
Inactive	10

Key Finding — Children Reduce Spending: Customers without any children at home spend an average of $1,106. Customers with children spend only $406 on average. This is a 2.7 times difference showing that family obligations significantly reduce discretionary spending.

```
-- Category-wise count
SELECT Spender_Category, COUNT(*) AS count
FROM spending_profile
GROUP BY Spender_Category;
```

Spender_Category	count
Medium Spender	749
Low Spender	1245
High Spender	246

Figure 14: Spender category distribution — 1,245 low spenders vs 246 high spenders

```
-- Shopping preference distribution
SELECT Shopping_Preference, COUNT(*) AS count
FROM channel_usage
GROUP BY Shopping_Preference;
```

Shopping_Preference	count
In-Store Shopper	1563
Mixed Shopper	297
Digital Shopper	370
Inactive	10

Figure 15: Shopping preference — 1,563 in-store shoppers dominate

```
SELECT * FROM spending_profile LIMIT 10;
```

Id	MntWines	MntFruits	MntMeatProducts	MntFishProducts	MntSweetProducts	MntGoldProducts	Total_Spend	Spender_Category	Category_Breadth
0	239	10	554	254	87	54	1198	Medium Spender	6
1	464	5	64	7	0	37	577	Medium Spender	5
9	57	0	27	0	0	36	120	Low Spender	3
13	19	0	5	0	0	8	32	Low Spender	3
17	637	47	237	12	19	76	1028	Medium Spender	6

Figure 16: Sample rows from spending_profile view showing category breadth data

STATISTICAL ANALYSIS

Three types of statistical tests were performed to validate findings with mathematical evidence.

Test 1 — T-Test: Do children at home affect spending and membership acceptance?

A T-Test compares the average values of two groups to see if the difference is real or just by chance.

Result: Customers with children spend $406 on average versus $1,106 for customers without children. The p-value is almost zero (2.02e-158) which means this difference is 100% statistically significant and not by chance. Similarly, the membership acceptance rate for customers with children is only 10.3% versus 26.5% for those without children (p-value = 1.04e-22).

Test 2 — ANOVA: Does education level affect total spending?

ANOVA tests whether differences across multiple groups are significant.

Result: Basic education customers spend $409, Graduation customers spend $620, Master's customers spend $612, and PhD customers spend $672. The p-value of 0.00000018 confirms that education level has a significant effect on spending.

Test 3 — Chi-Square: Is education level linked to membership acceptance?

Chi-Square tests whether two categorical variables are related.

Result: The p-value of 0.000088 confirms that education level is significantly related to whether a customer accepts the Gold Membership. PhD holders accept at 20.78% rate while Basic education customers accept at only 9.34%.

```
Acceptance Rate (With Kids): 0.10299625468164794
Acceptance Rate (No Kids): 0.26489028213166144
P-value (Acceptance): 1.0424367047639332e-22
```

Figure17: T-test results comparing spend and acceptance rate for families vs non-families

```
Chi-square p-value: 8.840050548990529e-05
```

Figure 18: Chi-square test confirming education and response are significantly related

Education	Response Rate	Average of Total_Spend
Basic	9.34%	409.39
Master	15.41%	611.78
Graduation	13.49%	619.90
PhD	20.78%	672.41
Total	14.91%	605.80

Figure19: Response rate and average spend by education level from Power BI

MACHINE LEARNING MODEL

A Random Forest Classification model was built to predict which customers will accept the Gold Membership offer. Random Forest works by building many decision trees and combining their results for a more accurate prediction.

Model Setup:

- Target variable: Response column (1 = accepted membership, 0 = rejected)
- Features used: Income, Age, Total Spend, NumWebPurchases, Recency
- Training data: 70% of customers (1,568 customers)
- Testing data: 30% of customers (672 customers)

Model Performance:

Metric	Value
Overall Accuracy	85.86%
Precision for Non-Acceptors	88%
Recall for Non-Acceptors	96%
Precision for Acceptors	52%
Recall for Acceptors	26%

The model is very good at identifying customers who will NOT accept (96% recall). However it struggles to identify all customers who WILL accept (only 26% recall). This is because the dataset is imbalanced — only 14.91% accepted, so the model sees far fewer examples of acceptors during training. This can be improved in future by using techniques like SMOTE or class weighting.

Top Features Driving Membership Acceptance:

Total Spend is the number one predictor — customers who already spend a lot are most likely to accept. Income is second. Recency is third — recent purchasers are more likely to accept. Age and NumWebPurchases also contribute.

```
Model Accuracy: 85.86%

Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.96      0.92       575
           1       0.52      0.26      0.34        97

    accuracy                           0.86       672
   macro avg       0.70      0.61      0.63       672
weighted avg       0.83      0.86      0.84       672
```

Figure 20: Random Forest model accuracy 85.86% with full classification report

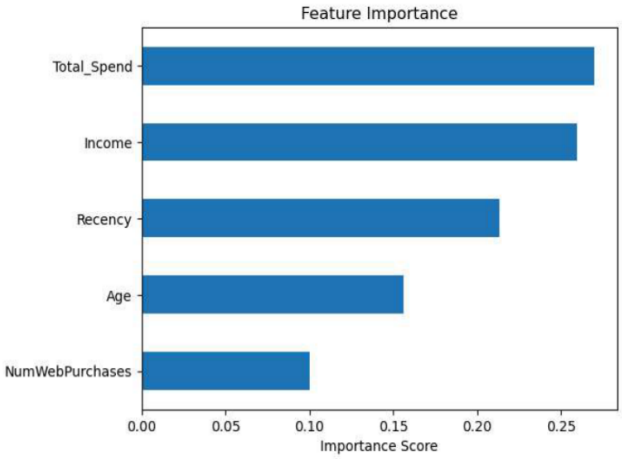

Figure21: Feature Importance chart — Total Spend and Income are the top predictors

MACHINE LEARNING MODEL — SMOTE IMPROVEMENT

The original model had 85.86% accuracy but dataset was imbalanced (85% vs 15%). Due to this, model was only finding 26 actual buyers out of 100.

Table 1 — Model Comparison:

Model	Accuracy	Buyers Found
Original	85.86%	26/100
Class Weight	85.86%	23/100
SMOTE	**77%**	**44/100**

SMOTE was applied on training data only. It created synthetic samples for minority class — balancing from 237 acceptors to 1,331.

Table 2 — Before vs After:

	Before SMOTE	After SMOTE
Class 0	1,331	1,331
Class 1	237	1,331

For a limited budget marketing campaign, SMOTE is the best model. It finds 44 buyers per 100 instead of 26 — saving cost and improving campaign results.

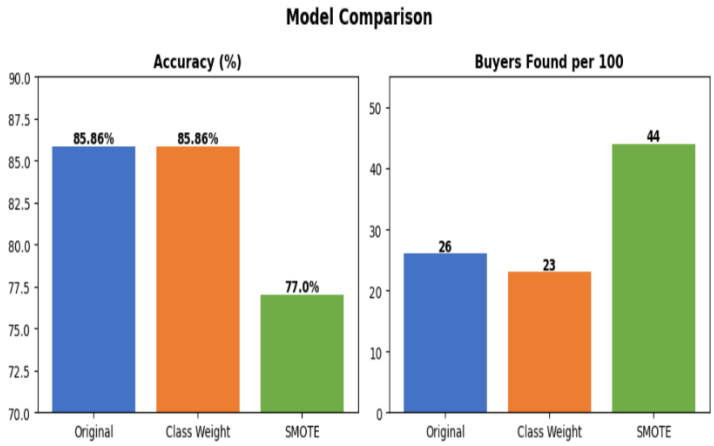

Figure 22: Model Comparison — SMOTE improves buyer recall from 26% to 44%

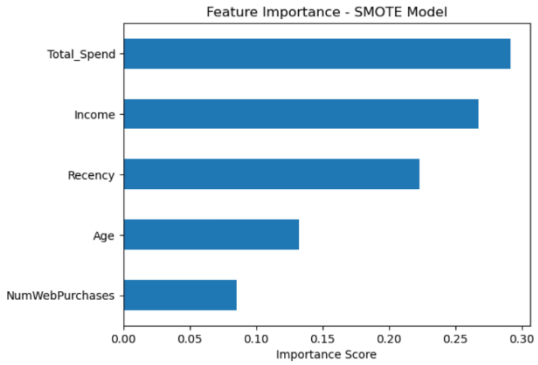

Figure 23: Feature Importance — Total Spend remains most important after SMOTE

STRATEGIC INSIGHTS — KEY FINDINGS

This section summarizes the most important patterns discovered across all stages of analysis.

Finding 1 — PhD customers are the best target segment PhD holders have the highest membership acceptance rate of 20.78% and spend $672 on average — the highest of any education group. Basic education customers respond at only 9.34%. This is a statistically confirmed finding (chi-square p-value = 0.000088).

Finding 2 — Families with children are the most resistant segment Customers with children at home have only 10.3% acceptance rate versus 26.5% for those without children. Their spending is also 2.7 times lower. The Gold Membership in its current form does not appeal to this group.

Finding 3 — In-store shoppers are the largest but least digitally engaged group 1,563 customers prefer physical stores. Only 370 are digital shoppers. Yet digital behavior (NumWebPurchases) is the 5th most important predictor of membership acceptance. Converting some in-store customers to digital could improve campaign results.

Finding 4 — Wine and Meat drive most revenue These two categories together account for approximately 77% of all customer spending. The Gold Membership offer should highlight benefits related to these categories.

Finding 5 — Low Spenders are the majority but give the least return 1,245 customers (55.6%) are Low Spenders. Trying to convert all of them is expensive. Resources should focus on Medium and High Spenders first.

Finding 6 — Recency is critical Customers who purchased recently are significantly more likely to accept the membership. Recency is the 3rd most important predictor in the model. Waiting too long to contact customers after their last purchase reduces conversion chances.

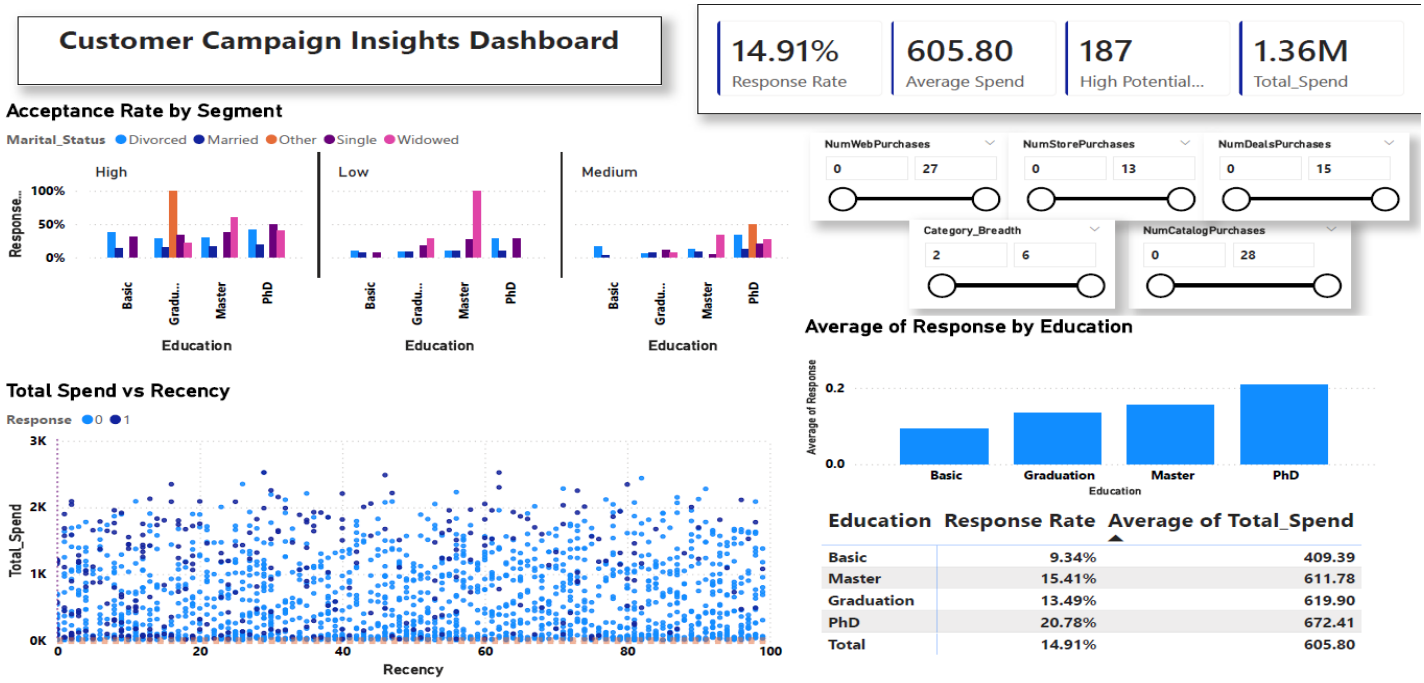

Figure 24: Customer Campaign Insights Dashboard showing acceptance rates and high potential segments

ACTIONABLE RECOMMENDATIONS

Recommendation 1 — Priority Email Campaign for PhD and Master's Degree Customers

Target the 856 PhD and Master's degree customers with a personalized email campaign offering early Gold Membership access. These customers have the highest acceptance rates of 20.78% and 15.41% and the highest average spending. Even a 5% increase in their acceptance rate would add approximately 43 new memberships from this group alone.

Recommendation 2 — Create a Family Gold Bundle

The current Gold Membership has very low appeal for families. Design a Family Gold Pack that includes discounts on food categories like Meat and Fruits which are more relevant to family households. This addresses the 10.3% acceptance rate gap versus 26.5% for non-families.

Recommendation 3 — Win-Back Campaign for High-Recency Customers

Customers who have not purchased recently are less likely to accept the membership. Identify customers with Recency above 60 days and send them a special catalogue offer or a limited-time reactivation discount. Recency is the 3rd most important predictor in the model so addressing it has a direct impact on response rates.

Recommendation 4 — Digital Conversion Initiative for In-Store Shoppers

Offer in-store customers a bonus incentive to try the website — for example a 10% first online purchase discount. This builds digital behavior which is linked to higher membership acceptance rates. The 1,563 in-store shoppers represent a large untapped opportunity.

Recommendation 5 — Wine and Meat Membership Bundle

Design the Gold Membership value proposition around Wine and Meat since these are the categories customers already love. Position the membership as a "Wine and Gourmet Food Club" rather than a generic discount card. This creates stronger emotional connection to the offer.

Recommendation 6 — VIP Outreach for High Spenders

The 246 High Spenders (11% of base) are the most likely to accept the membership based on the model. Give them VIP treatment — personal phone calls, exclusive preview access, or additional benefits like free delivery. Total Spend is the number one predictor of acceptance, so this group should always be the first priority.

CONCLUSION

This project analyzed 2,240 Superstore customers across all major data tools — Excel for validation, MySQL for database management, Python for analysis and machine learning, and Power BI for dashboarding. The overall Gold Membership acceptance rate of 14.91% indicates that while the campaign reaches a large audience, it is only converting a small portion of customers.

The Random Forest model achieved 85.86% accuracy and confirmed that Total Spend, Income, and Recency are the three most powerful predictors of membership acceptance. Statistical tests confirmed that education level and presence of children at home significantly affect both spending and acceptance behavior.

By targeting PhD-educated customers, creating family-friendly membership tiers, re-engaging recently inactive customers, and building digital shopping habits, the Superstore can meaningfully increase its Gold Membership uptake and maximize the return on its marketing investment.

To further improve campaign targeting, three machine learning models were tested to address the imbalanced dataset. The original Random Forest model could only identify 26 buyers per 100 customers. Class weight balancing did not help — recall dropped to 23. After applying SMOTE, buyer identification improved to 44 per 100 customers. Total Spend remained the most important factor across all three models, confirming it is a reliable predictor. SMOTE is recommended for future campaigns as it finds more real buyers without increasing budget.

TOOLS AND TECHNOLOGIES USED

Tool	Purpose
Microsoft Excel	Data validation, conditional formatting, anomaly detection
MySQL Workbench	Database creation, DDL scripts, views, sanity queries
Python (Jupyter Notebook)	Data cleaning, EDA, feature engineering, statistical tests, ML model
Power BI	Executive dashboard and Campaign insights dashboard
Pandas and NumPy	Data manipulation
Matplotlib and Seaborn	Charts and visualizations
Scikit-learn	Random Forest model, train-test split, classification report
SQLAlchemy	Python to MySQL database connection